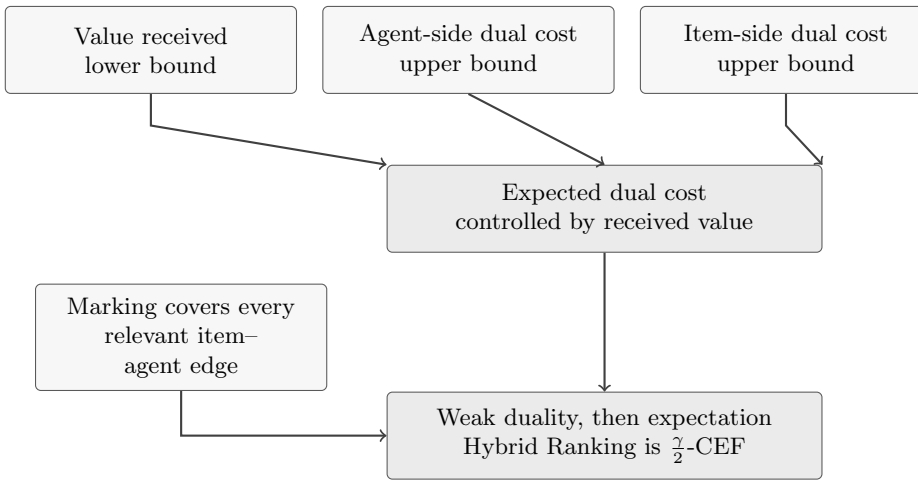


Schematic proof dependency diagram. An arrow means that the input is used to prove the output. For $\gamma > 0$, the three conditional increment bounds imply that $\frac{\gamma}{2}$ times the expected dual cost is at most the expected value received by class i. The marking construction is feasible for every realization, so weak duality applies before taking expectations. The case $\gamma = 0$ is the trivial zero-CEF endpoint. Boxes show proof obligations, not stages of the online algorithm.